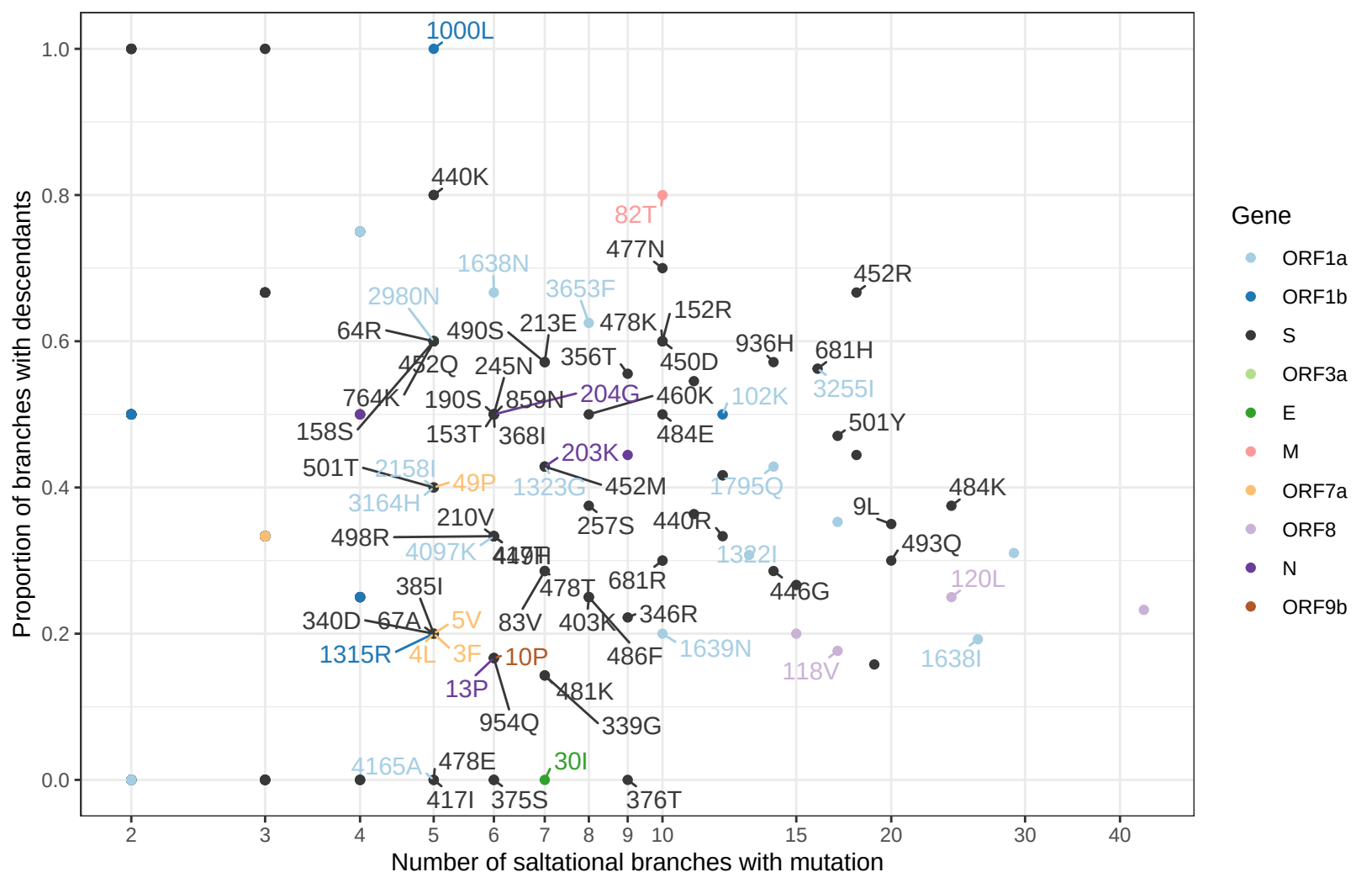
Proportion of branches with descendants
Number of saltational branches with mutation
1.0
0.8
0.6
0.4
0.2
0.0
2
3
4
5
6
7
8
9
10
15
20
30
40
1000L
440K
82T
477N
1638N
452R
2980N
3653F
152R
64R
490S
213E
478K
936H
681H
452Q
245N
356T
450D
3255I
764K
190S
859N
204G
460K
102K
501Y
158S
153T
368I
484E
203K
501T
2158I
49P
1323G
452M
1795Q
484K
3164H
9L
210V
257S
440R
498R
493Q
4097K
1322I
120L
385I
681R
446G
340D
67A
5V
83V
403K
346R
1315R
4L
3F
10P
486F
1639N
118V
1638I
13P
481K
954Q
339G
4165A
478E
30I
417I
375S
376T
Gene
ORF1a
ORF1b
S
ORF3a
E
M
ORF7a
ORF8
N
ORF9b